

Assignment 4

Binary Classification with 1D CNN

Appliance Energy Prediction Dataset

Course: AI & Neural Networks

University: Astana IT University (AITU)

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1. Introduction

This report presents the implementation and evaluation of a 1D Convolutional Neural Network (1D CNN) for a binary classification task on the Appliance Energy Prediction dataset. The dataset contains 19,735 observations sampled every 10 minutes over about 4.5 months, covering temperature and humidity readings across nine rooms plus outdoor weather variables.

The target variable is created by thresholding appliance energy consumption at its median value: $\text{HighConsumption} = 1$ when Appliances > median (60 Wh), otherwise 0. The CNN is compared against a logistic regression baseline trained on the same rolling-window representation.

2. Data Preparation

2.1 Dataset

The dataset `energydata_complete.csv` contains 19,735 rows and 29 columns. Features include indoor temperatures (T1-T9), indoor humidities (RH_1-RH_9), outdoor weather (T_out, Press_mm_hg, RH_out, Windspeed, Visibility, Tdewpoint), and two random reference variables (rv1, rv2). Four cyclical time features were engineered from the timestamp: `hour_sin`, `hour_cos`, `dow_sin`, `dow_cos`. Total input features: 32.

2.2 Target Variable

The binary target `HighConsumption` was created using the median appliance consumption (60 Wh). The class distribution is close to balanced, with 45.6% high-consumption and 54.4% low/normal samples. Because the imbalance gap is well below 20%, class weighting was not applied during training.



Figure 1: Target Class Distribution

2.3 Chronological Split and Windowing

The data was split in a 70/15/15 ratio preserving chronological order. Rolling windows of 24 consecutive time steps (4 hours of history at 10-minute intervals) were constructed so that each sample has shape (24, 32). StandardScaler was fit on the training set only and applied to validation and test sets to prevent data leakage.

- Train sequences: (13790, 24, 32)
- Validation sequences: (2936, 24, 32)
- Test sequences: (2937, 24, 32)

3. 1D CNN Model

3.1 Architecture

A compact 1D CNN was used, treating each 24-step window as a 1D signal over 32 channels:

Input (24, 32) -> Conv1D(filters, kernel, causal, ReLU) -> MaxPool1D(2) -> Flatten -> Dense(64, ReLU) -> Dropout(0.3) -> Dense(1, Sigmoid)

Causal padding ensures the convolution at position t only uses steps $\leq t$, keeping the temporal order intact. Binary cross-entropy loss and the Adam optimizer (learning rate $1e-3$) were used, with EarlyStopping (patience 4 on val_loss, restore_best_weights=True) and ModelCheckpoint saving the best epoch. Training used batch size 64 with a maximum of 20 epochs.

3.2 Hyperparameter Search

Two CNN configurations were compared on the validation set:

filters	kernel	Accuracy	Precision	Recall	F1
64	3	0.8229	0.8539	0.7557	0.8018
128	5	0.7960	0.8015	0.7572	0.7787

Config A (filters=64, kernel=3) achieved the higher validation F1 (0.802 vs 0.779) and was selected as the best model. It was saved to cnn_classifier.h5.

3.3 Training Curves

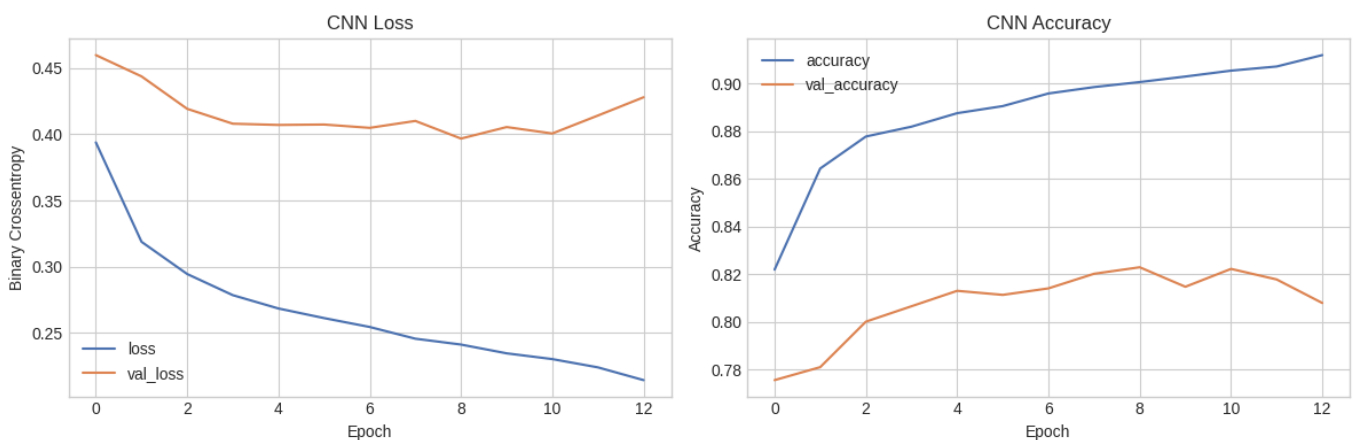


Figure 2: CNN Training vs Validation Loss and Accuracy

4. Evaluation

4.1 Test Metrics

Model	Accuracy	Precision	Recall	F1
1D CNN (best)	0.7995	0.7687	0.8498	0.8072
Logistic Regression	0.8383	0.8223	0.8580	0.8398

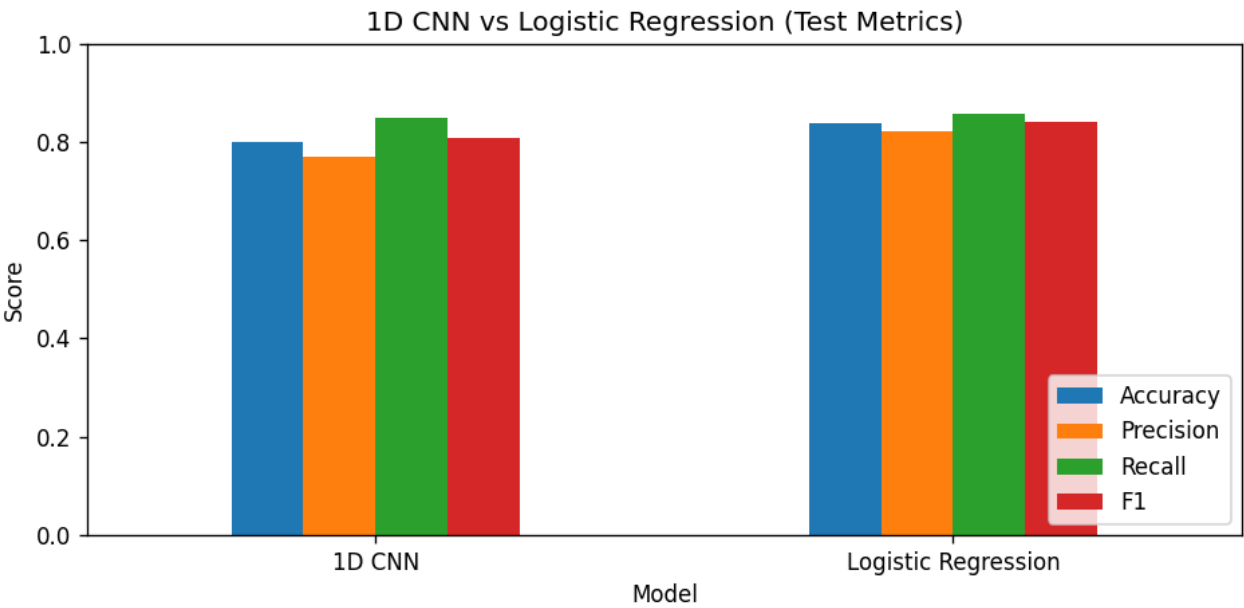


Figure 3: CNN vs Logistic Regression (Test Metrics)

4.2 Confusion Matrix

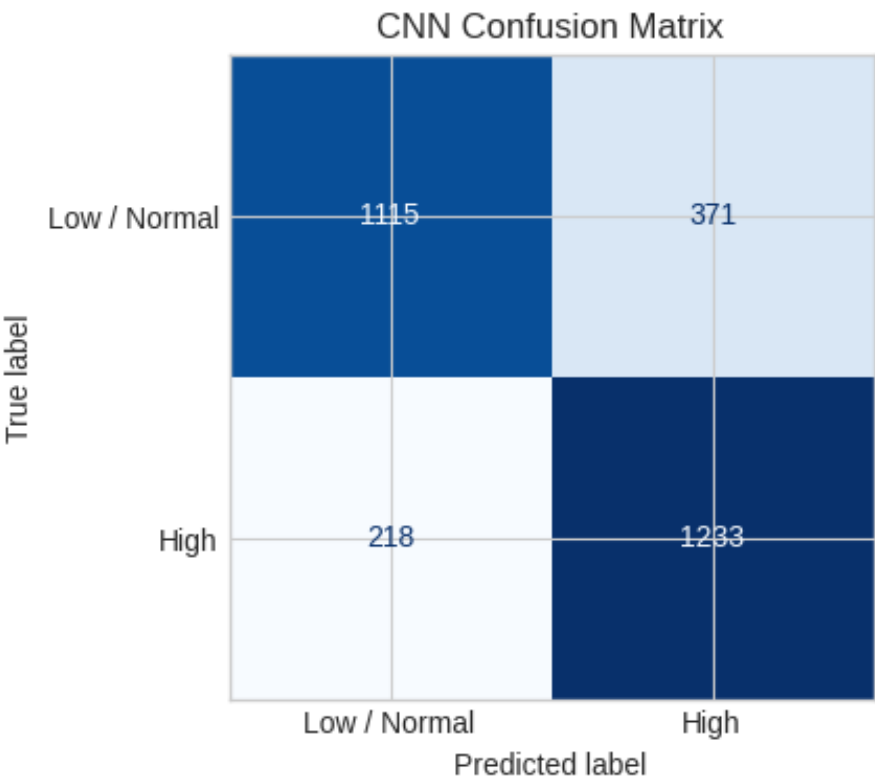


Figure 4: CNN Confusion Matrix on Test Set

5. Discussion

The 1D CNN reached reasonable test performance (Accuracy 0.80, F1 0.81). The smaller config (filters=64, kernel=3) generalized better than the larger one (filters=128, kernel=5), suggesting that a wider receptive field and more filters caused mild overfitting on this dataset of modest size.

The logistic regression baseline, fit on the flattened 24x32 window (768 features), is competitive with - and in this run slightly stronger than - the CNN (F1 0.840 vs 0.807). This suggests that a large part of the signal separating high vs normal consumption is captured by simple linear combinations of the scaled features, especially the cyclical time features (hour_sin, hour_cos) which correlate strongly with daily usage patterns.

The confusion matrix shows the CNN favors recall over precision on the High class (recall 0.85 vs precision 0.77): it catches most high-consumption windows but also misclassifies a noticeable share of normal windows as high. A threshold above 0.5 would trade recall for precision if false positives are more costly.

Causal padding was important here: without it the model would have access to future time steps within a window. Keeping the split strictly chronological and fitting the scaler only on the training set prevents the model from peeking into the future.

6. Conclusions

1. A binary classification task was successfully formulated from the raw Appliance Energy dataset by thresholding at the median consumption (60 Wh).
2. A 1D CNN with causal Conv1D layers was trained on 24-step windows of 32 features and achieved Accuracy = 0.7995 and F1 = 0.8072 on the held-out test set.
3. A logistic regression baseline on the same flattened windows reached F1 = 0.8398, showing that the dataset has a strong linear component and that a heavier CNN is not automatically better for tabular-style short windows.
4. For further gains on this problem, richer temporal architectures (stacked Conv1D, dilated convolutions, LSTM/GRU, Transformers) and class-balanced thresholds beyond 0.5 would be natural next steps.
5. The best CNN model was saved as `cnn_classifier.h5` in the Assignment4 directory.